

Technical Publication

Diesel Engine
8V 396 TE74L

Description and
Operating Instructions
M010995/00E

DETROIT DIESEL



DaimlerChrysler Off-Highway

Printed in Germany

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Il manuale va consultato per evitare anomalie o guasti durante il servizio, per cui va messo a disposizione dall'utente al personale addetto alla manutenzione e alla condotta.

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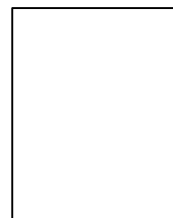
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M010995/00E

**Description and
Operation Manual**

8V 396 TE74L



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88040 Friedrichshafen
GERMANY

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Notes for the user

This Manual consists of 10 Sections, marked by tab-index sheets.

These Sections are listed on this page.

Each Section has its own detailed Table of Contents which lists the Groups and Subgroups.

The Groups in Sections B and G correspond to the Groups in the Spare Parts Catalog and other Product Support publications.

Using the Table of Contents to trace an entry

- ➡ Table of Contents (general)
- ➡ Table of Contents (section)
- ➡ Group / Subgroup
- ➡ Heading

Using the Alphabetical Index to trace an entry

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Publication Amendment Service

Please complete and return the Confirmation of Receipt card; you will then receive amendments to your manual.

A	Product Summary Engine summary, technical data
B	Description System and subassembly descriptions
C	Operating Instructions Task descriptions
D	Maintenance Schedule Schedule and maintenance frequency chart
E	Troubleshooting Fault conditions, probable causes, remedies
F	Drawings Engine cross-sections
G	Task Description Inspection and maintenance
K	Preservation Engine preservation
Z	Appendix Supplementary documentation
?	Index Alphabetical index of key



Contents

A.000 Product summary - Group A

A.010 Engine summary
Engine model designation
Engine power
Engine layout - cutaway drawings

A.020 Engine side and cylinder designations
Designations and abbreviations

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A.040 Main engine dimensions
Engine weight, coolant and oil capacities

A.050 Operational data
Pressures
Temperatures
Consumptions



Diesel Engine**Part A**
Product Summary

The product summary gives details of the engine and contains technical data such as engine power, general engine specifications, engine weight, oil and coolant capacities, settings and operational data.

ENGINE LAYOUTKey to Engine Model Designation 8 V 396 TE 74L

- 8 = number of cylinders
V = Vee configuration
396 = engine series - 100 times displacement of one cylinder in liters
T = exhaust turbocharging
E = charge air-to-coolant cooling in split-circuit coolant system
7 = marine engine
4 = design digit
L = uprated

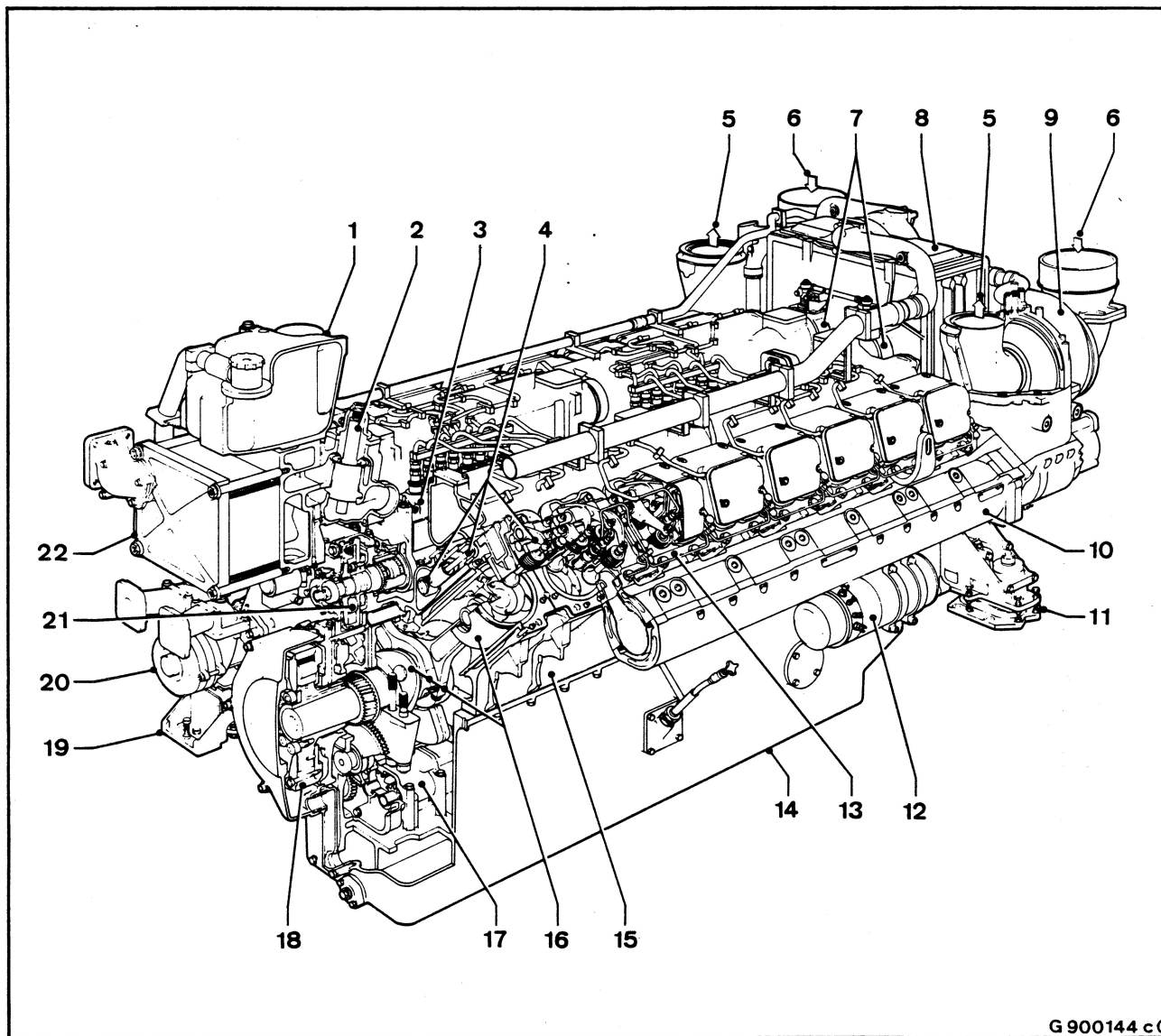
Engine Power

Continuous power 1000 kW at 2000 rpm (fuel stop power)

Reference conditions

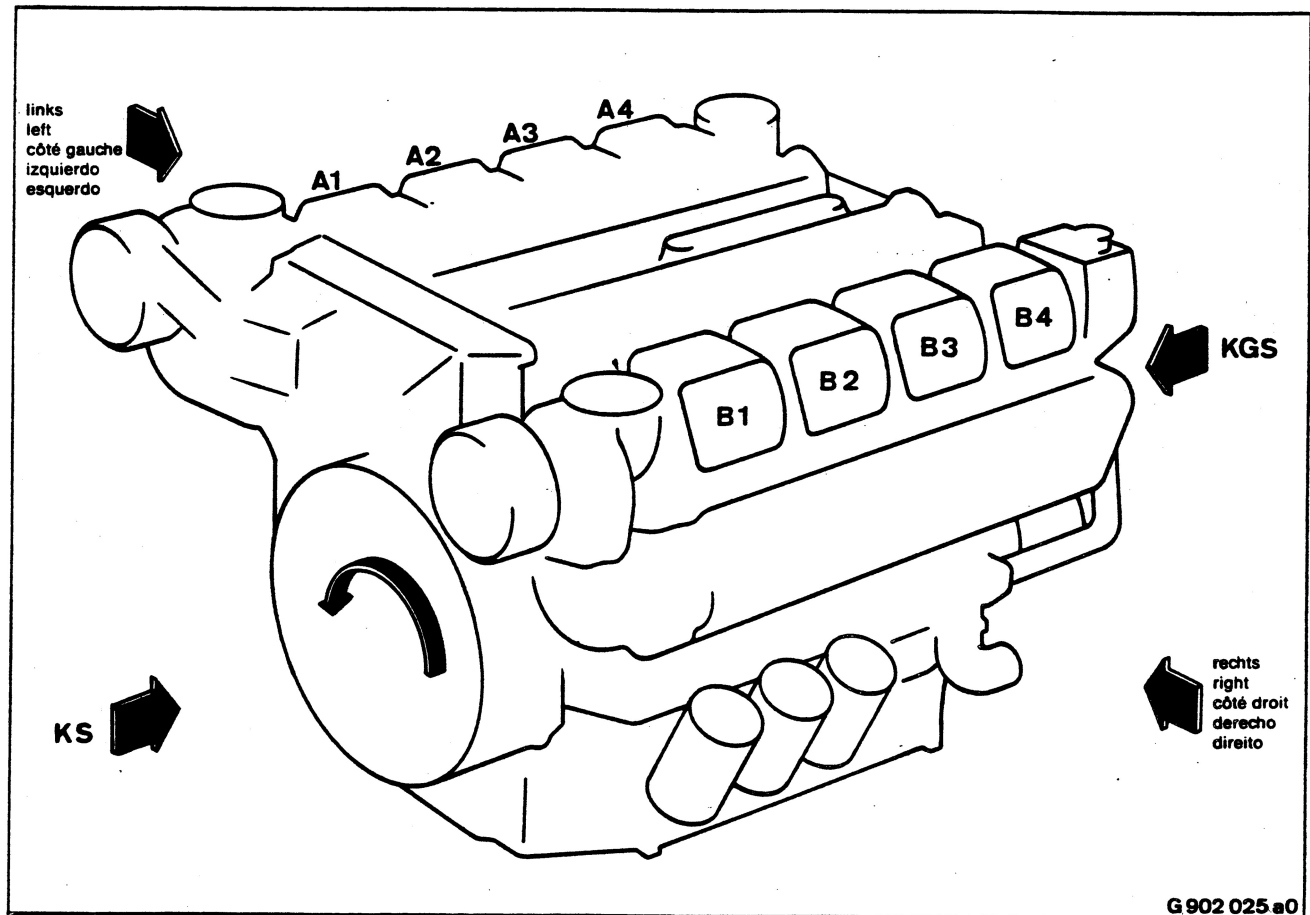
- intake air temperature 25°C
- raw water temperature 25°C
- barometric pressure 1000 mbar

Engine Layout Cutaway Drawing 16V 396 TE.4 (also applies to 8 and 12V)



G 900144 c0

- | | |
|-------------------------------|-----------------------------|
| 1 Coolant expansion tank | 12 Starter |
| 2 Coolant thermostat | 13 Cylinder head |
| 3 Fuel injection pump | 14 Oil pan |
| 4 Valve gear | 15 Crankcase |
| 5 Exhaust outlet | 16 Running gear |
| 6 Air inlet | 17 Engine oil pump |
| 7 Emergency air shut-off flap | 18 Vibration damper |
| 8 Intercooler | 19 Engine mount, timing end |
| 9 Exhaust turbocharger | 20 Bilge pump |
| 10 Exhaust manifold | 21 Fuel injection timer |
| 11 Engine mount, flywheel end | 22 Coolant cooler |

ENGINE SIDE AND CYLINDER DESIGNATIONS

Engine sides are always designated as viewed from the flywheel end.

The left bank of cylinders is marked "A" and the right bank "B" (according to DIN standard 6265).

The cylinders of each bank are numbered consecutively, starting with No. 1 at the flywheel end.

Numbering of other engine components is also from the flywheel end, starting with No. 1.

Designations and abbreviations

Flywheel end = KS (formerly HKS)

Timing end = KGS (formerly GKS)

Left side (formerly A side)

Right side (formerly B side)

GENERAL SPECIFICATIONS

Operating method	four-stroke cycle, single-acting
Combustion method	direct injection
Mode of supercharging	single-stage sequential turbocharging
Cooling method	liquid-cooled
Cylinder arrangement	90° Vee
Bore	165 mm
Stroke	185 mm
Displacement, cylinder	3.96 liters
Number of cylinders	8
Displacement, total	31.6 liters
Compression ratio	12.3 : 1
Direction of rotation (viewed from flywheel end)	c.c.w. (non-reversible)
Firing order	A1 A4-A2 A3 B4 B3 B2-B1
Fuel injection pressure:	
Setting pressure with new spring	280 to 288 bar
Specified pressure with run-in spring	270 to 278 bar
Minimum pressure with run-in spring	260 bar
Final compression pressure at 120 rpm, 1000 mbar and engine coolant at 5°C	20 to 24 bar
Break-away torque at engine coolant temperature of 40°C, allowance for acceleration included	approx. 400 Nm
Cranking torque at engine coolant temperature of 40°C	approx. 370 Nm
Firing speed at engine coolant temperature of 40°C	approx. 120 rpm
Mean piston speed at 2000 rpm	12.33 m/s